Oasistek®

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SMD LED
LAMP LED
HIGH POWER

Product Catalogue

Taiwan Oasis Technology Co., Ltd.
李洲科技股份有限公司

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SMD LED Lamp LED HIGH POWER Product Catalogue

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SMD LED

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Lighting
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TO-3806 15

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TO-5050 18

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HIGH POWER
TOP
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TOP-BLUE 23
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Lamp LED

3mm Series
3mm Round
TOL-30 26
3mm Concave & Oval
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5mm Series
5mm Round
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Oasistek® Taiwan Oasis Technology Co., Ltd.

Introduction

Taiwan Oasis Technology Co., Ltd., was established by Mr. Ming-Shun Lee in 1974 with an ideal to provide the marketplace simplicity, integrity reliability, progress, precision, and innovation.

The company started its life initially as a dedicated downstream LED technology manufacturing service focused on items such as plastic molding, injection and packing.

As Oasistek expanded its business vertically and upgraded its competencies, it soon began to move into producing a complete range of products, such as High Power LEDs, LAMP LEDs, SMD LEDs, LED Displays and Lighting. Additionally, Oasistek also offers customization manufacturing services to help customers' meet their sales targets.

InLED recent years, Oasistek has proudly introduced its LED Video Display and Lumichain™ series to the marketplace. Oasistek has been applauded globally and is reputed as a pioneer in Taiwan for being the 1st Taiwanese LED manufacturer to consistently supply the Japanese market.

Today, Oasistek's complete product line-up is now well represented worldwide and is globally known for their quality and cutting-edge technology. Delivering a full range of services and meeting customers' need is its mission and Oasistek is sincerely inviting you to be a partner and step forward together.

Taipei Headquarter

Taoyuan Manufacturing Center

Dongguan Manufacturing Center

Milestones

- 2016 The New headquarter has been relocated to U-Town Building in Xizhi Dist., New Taipei City.
- 2015 All new LED Spot Light product has been published.
- 2013 The winner of Transformation and Upgrading of the processing Trade Industry Award held in Dongguan China, power was confirmed.
- 2012 Increase the new production line, the production capacity increased by more then 40KK/M. Establishment of an independent R&D department, the main R&D direction of illumination and backlight. Set up offices in Shenzhen and Qingdao and build up a unique channel marketing model.
- 2011 Launched a series of interior commercial lighting products.
- 2010 Launched "Star Vessel" which is functional and designing outdoor street light.
- 2009 Adopted a new brand identity "Oasistek" and a renewed spirit.
- 2008 China Hwa-Tai Factory completed. Adopted Green Policy. Lumichain trademark registered
- 2007 Began offering full product range from LEDs components to LED Video Displays.
- 2005 Announced partnership with leading Japanese corporation, White LED Technology. Lumichain series launched.
- 2004 Publicly Listed on the Taiwan Stock Exchange. ISO/TS 16949 Certified.
- 2003 Inaugurated fully automated line for LED Display production.
- 2001 Renamed to Taiwan Oasis Technology Co., Ltd. QS 9000 Certificated.
- 1999 Began SMD LED mass production. China factories ISO 9002 certified.
- 1998 Approval of Stock Issuance. Patent awarded for LED Packaging technology. Taiwan factories ISO 9002 certified
- 1996 Kwei Shan, Taiwan factory completed. Completed LED products launched.
- 1994 Merged with Lihsin Co., Ltd. Patent awarded for advanced CD package design.
- 1990 Dong-Guan, China factory completed. Patent awarded for assembling exchangeable plastic injection mold structures.
- 1989 Chung-Ho, Taiwan factory completed. Manufacturing of LED Indicators and LED Displays.
- 1984 Nan-Kan, Taiwan No.2 completed. PCB Manufacturing.
- 1980 Nan-Kan, Taiwan Factory No.1 completed. Manufacturing of Plastic Molds, Plastic Injections and, Reflectors for LED.
- 1974 Taiwan Oasis Enterprise Co., Ltd founded.

Product Line

1 LED Component Group

- LED Lamps
- SMD LED
- Holder Lamp
- LED Display
  - Seven Segment
  - Alphanumeric
  - Light Bar
  - Dot Matrix

2 Lighting Module Group

- Lamp / Display Module
- Backlight Module
- Traffic Signal Module
- Street Lamp Module

3 System Group

- LED Cluster

4 Extended Customization Group

- Mold
- Plastic
  - Reflector
  - Light-guide Plate
  - Backlight Board
  - LED Through Holder
  - LED Cover Part
- PCB

SMD LED

TO-2835SWNQ-H1-E

1 | 2 | 3 | 4 | 5 | 6

1 Taiwan Oasistek

2 Illumination Color Code

SW	5000-9500K
FW	3500-5000K
TW	2600-3500K

3 CRI Index

N	Ra≥80
M	Ra≥75
L	Ra≥70

4 Viewing Angel

N	20°
O	60°
P	90°
Q	120°
R	140°
S	175°
X	None

5 Packaging Lens Type

C	Small Size Chip
D	Small Size Chip
E	Medium-sized Chip
F	Medium-sized Chip
H	Medium-power Chip

6 Special Type (Option)

B	Rubbing-up series (Black surface)
BU	Reverse Packaging series
E	Add antistatic Zener
A	Triple crystal white lightmaterial, Anode connected and Cathode seperated
N	Reversed polarity material (chip)

Bi-Color

TO-3228BC-MRMGFF

1 | 2 | 3 | 4 | 5 | 5 | 6 | 6

Tri-Color

TO-3228BC-MRPBFGF

1 | 2 | 3 | 4 | 5 | 5 | 6 | 6 | 6

1 Taiwan Oasistek

2 Package Dimension and Type

Code	Dimension and Lens Type
3228	[3.2mm x 2.8mm] Frame Type
3020	[3.0mm x 2.0mm] Frame Type
5050	[5.0mm x 5.0mm] Frame Type
2808	[2.8mm x 0.8mm] Side Emitting
3806	[3.8mm x 0.6mm] Side Emitting

3 Lens Type

Code	Lens Type
A	Dome
B	Plane
S	Side View
T	Thin Thickness (0.4mm for 1608 Package)
D	Special Thickness (0.65mm)
K	Black Plane

4 Lens Color

Code	Lens Color
C	Water Clear
D	Color Diffused
R	Red Diffused (Fluorescent)
G	Green Diffused (Fluorescent)
Y	Yellow Diffused (Fluorescent)
W	White Diffused
T	Color Transparent
F	Use Phosphors

5 Emitting Color Code

Single chip

Code	Color & Material	Wavelength (nm)
V	Violet (GaN/SiC)	405
B	Blue (InGaN/GaN)	468
BG	Bluish Green(InGaN)	505
P	Pure Green (InGaN/GaN)	525
G	Yellow Green (GaP)	570
MG	High Bright Green (AlGaInP)	572
Y	Yellow (GaAsP/GaP)	585
MY	High Bright Yellow (AlGaInP)	592
MA	High Bright Amber (AlGaInP)	605
ME	High Bright Orange(AlGaInP)	623
E	Orange (GaAsP/GaP)	630
MR	High Bright Red (AlGaInP)	630
MRL	High Bright Red (AlGaInP)	640
R	Red (GaP)	700

Color Temperature : 3000~3500 (eg.TO-3228BY-MWaK)

Multi-Chip Combination of above chip code

6 Brightness Code

Code	Typical Luminous Intensity (mcd)
A	0.1~2.2
B	2.2~6.5
C	6.5~18.0
D	18.0~51.0
E	51.0~146.0
F	146.0~417.0
G	417.0~917.0
H	917.0~2015.0
K	2015.0~3406.0
M	3406.0~5756.0

TO-B0603BC-MR-E

1 2 3 4 5 6 7

1 Taiwan Oasistek

2 PCB Type Code Omitted if Normal

A	Slot from 0.35mm
B	Slot from 0.55mm
D	Slot from 1.10mm

3 Package Dimension and Type

Code	Dimension and Lens Type
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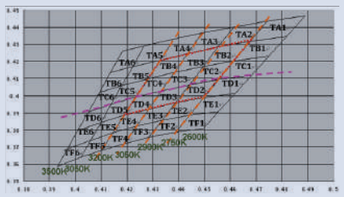
B	2.2~6.5
C	6.5~18.0
D	18.0~51.0
E	51.0~146.0
F	146.0~417.0
G	417.0~917.0
H	917.0~2015.0
K	2015.0~3406.0
M	3406.0~5756.0

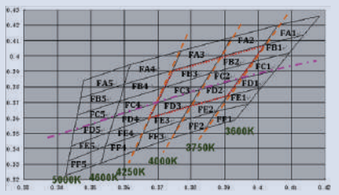
Color Temperature & TW



Note: ⬚ is high bin

TA1					
X	0.4889	0.4742	0.4677	0.4817	0.4889
Y	0.4466	0.4436	0.4319	0.4347	0.4466
TA2					
X	0.4742	0.4613	0.4553	0.4677	0.4742
Y	0.4436	0.44	0.4284	0.4319	0.4436
TA3					
X	0.4613	0.4489	0.4435	0.4553	0.4613
Y	0.44	0.4367	0.4246	0.4284	0.44
TA4					
X	0.4489	0.438	0.433	0.4435	0.4489
Y	0.4367	0.433	0.4206	0.4246	0.4367
TA5					
X	0.438	0.4267	0.4223	0.433	0.438
Y	0.433	0.4293	0.4171	0.4206	0.433
TA6					
X	0.4267	0.4184	0.4137	0.4223	0.4267
Y	0.4293	0.4261	0.4138	0.4171	0.4293
TB1					
X	0.4817	0.4677	0.4615	0.4749	0.4817
Y	0.4347	0.4319	0.4205	0.4232	0.4347
TB2					
X	0.4677	0.4553	0.4496	0.4615	0.4677
Y	0.4319	0.4284	0.4171	0.4205	0.4319
TB3					
X	0.4553	0.4435	0.4383	0.4496	0.4553
Y	0.4284	0.4246	0.4136	0.4171	0.4284
TB4					
X	0.4435	0.433	0.4282	0.4383	0.4435
Y	0.4246	0.4206	0.4097	0.4136	0.4246
TB5					
X	0.433	0.4223	0.418	0.4282	0.433
Y	0.4206	0.4171	0.4059	0.4097	0.4206
TB6					
X	0.4223	0.4137	0.4093	0.418	0.4223
Y	0.4171	0.4138	0.402	0.4059	0.4171
TC1					
X	0.4749	0.4615	0.4556	0.4682	0.4749
Y	0.4232	0.4205	0.4095	0.412	0.4232
TC2					
X	0.4615	0.4496	0.444	0.4556	0.4615
Y	0.4205	0.4171	0.4063	0.4095	0.4205
TC3					
X	0.4496	0.4383	0.4334	0.444	0.4496
Y	0.4171	0.4136	0.403	0.4063	0.4171
TC4					
X	0.4383	0.4282	0.4235	0.4334	0.4383
Y	0.4136	0.4097	0.3993	0.403	0.4136
TC5					
X	0.4282	0.418	0.4139	0.4235	0.4282
Y	0.4097	0.4059	0.395	0.3993	0.4097
TC6					
X	0.418	0.4093	0.4051	0.4139	0.418
Y	0.4059	0.402	0.3906	0.395	0.4059
TD1					
X	0.4682	0.4556	0.4498	0.4618	0.4682
Y	0.412	0.4095	0.3989	0.4013	0.412
TD2					
X	0.4552	0.444	0.4387	0.4498	0.4552
Y	0.4095	0.4063	0.3958	0.3989	0.4095
TD3					
X	0.444	0.4334	0.4285	0.4387	0.444
Y	0.4063	0.403	0.3921	0.3958	0.4063
TD4					
X	0.4334	0.4235	0.419	0.4285	0.4334
Y	0.403	0.3993	0.3886	0.3921	0.403
TD5					
X	0.4235	0.4139	0.4099	0.419	0.4235
Y	0.3993	0.395	0.3843	0.3886	0.3993
TD6					
X	0.4139	0.4051	0.4009	0.4099	0.4139
Y	0.395	0.3906	0.3796	0.3843	0.395
TE1					
X	0.4618	0.4498	0.4442	0.4555	0.4618
Y	0.4013	0.3989	0.3886	0.3909	0.4013
TE2					
X	0.4498	0.4387	0.4335	0.4442	0.4498
Y	0.3989	0.3958	0.3857	0.3886	0.3989
TE3					
X	0.4387	0.4285	0.4238	0.4335	0.4387
Y	0.3958	0.3921	0.3822	0.3857	0.3958
TE4					
X	0.4285	0.419	0.4146	0.4238	0.4285
Y	0.3921	0.3886	0.3785	0.3822	0.3921
TE5					
X	0.419	0.4099	0.406	0.4146	0.419
Y	0.3886	0.3843	0.3739	0.3785	0.3886
TE6					
X	0.4099	0.4009	0.397	0.406	0.4099
Y	0.3843	0.3796	0.3689	0.3739	0.3843
TF1					
X	0.4555	0.4442	0.4388	0.4495	0.4555
Y	0.3909	0.3886	0.3787	0.3808	0.3909
TF2					
X	0.4335	0.4285	0.4388	0.4442	0.4335
Y	0.3857	0.3758	0.3787	0.3886	0.3857
TF3					
X	0.4335	0.4238	0.4193	0.4285	0.4335
Y	0.3857	0.3822	0.3721	0.3758	0.3857
TF4					
X	0.4238	0.4146	0.4103	0.4193	0.4238
Y	0.3822	0.3785	0.3682	0.3721	0.3822
TF5					
X	0.4146	0.406	0.4023	0.4103	0.4146
Y	0.3785	0.3739	0.3642	0.3682	0.3785
TF6					
X	0.406	0.397	0.3931	0.4023	0.406
Y	0.3739	0.3689	0.3587	0.3642	0.3739

Color Temperature & FW



Note: ⬚ is high bin

FA1					
X	0.4184	0.4058	0.4019	0.4137	0.4184
Y	0.4261	0.42	0.4074	0.4138	0.4261
FA2					
X	0.4058	0.392	0.3888	0.4019	0.4058
Y	0.42	0.4121	0.3996	0.4074	0.42
FA3					
X	0.392	0.377	0.3745	0.3888	0.392
Y	0.4121	0.4035	0.3909	0.3996	0.4121
FA4					
X	0.377	0.3618	0.3601	0.3745	0.377
Y	0.4035	0.3945	0.3818	0.3909	0.4035
FA5					
X	0.3618	0.3479	0.3469	0.3601	0.3618
Y	0.3945	0.3841	0.3728	0.3818	0.3945
FB1					
X	0.4137	0.4019	0.3981	0.4093	0.4137
Y	0.4138	0.4074	0.396	0.402	0.4138
FB2					
X	0.4019	0.3888	0.3857	0.3981	0.4019
Y	0.4074	0.3996	0.3887	0.396	0.4074
FB3					
X	0.3888	0.3745	0.3722	0.3857	0.3888
Y	0.3996	0.3909	0.3799	0.3887	0.3996
FB4					
X	0.3745	0.3601	0.3586	0.3722	0.3745
Y	0.3909	0.3818	0.371	0.3799	0.3909
FB5					
X	0.3601	0.3469	0.346	0.3586	0.3601
Y	0.3818	0.3728	0.362	0.371	0.3818
FC1					
X	0.4093	0.3981	0.3944	0.4051	0.4093
Y	0.402	0.396	0.385	0.3906	0.402
FC2					
X	0.3981	0.3857	0.3826	0.3944	0.3981
Y	0.396	0.3887	0.3781	0.385	0.396
FC3					
X	0.3857	0.3722	0.3699	0.3826	0.3857
Y	0.3887	0.3799	0.3699	0.3781	0.3887
FC4					
X	0.3722	0.3586	0.3572	0.3699	0.3722
Y	0.3799	0.371	0.3609	0.3699	0.3799
FC5					
X	0.3586	0.346	0.345	0.3572	0.3586
Y	0.371	0.362	0.3516	0.3609	0.371
FD1					
X	0.4051	0.3944	0.3909	0.4009	0.4051
Y	0.3906	0.385	0.3743	0.3796	0.3906
FD2					
X	0.3944	0.3826	0.3797	0.3909	0.3944
Y	0.385	0.3781	0.3679	0.3743	0.385
FD3					
X	0.3826	0.3699	0.3677	0.3797	0.3826
Y	0.3781	0.3699	0.3603	0.3679	0.3781
FD4					
X	0.3699	0.3572	0.3558	0.3677	0.3699
Y	0.3699	0.3609	0.351	0.3603	0.3699

FD5					
X	0.3572	0.345	0.3442	0.3558	0.3572
Y	0.3609	0.3516	0.3415	0.351	0.3609

FE1					
X	0.4009	0.3909	0.3875	0.397	0.4009
Y	0.3796	0.3743	0.3641	0.3689	0.3796

FE2					
X	0.3909	0.3797	0.3769	0.3875	0.3909
Y	0.3743	0.3679	0.3579	0.3641	0.3743

FE3					
X	0.3797	0.3677	0.3656	0.3769	0.3797
Y	0.3679	0.3603	0.35	0.3579	0.3679

FE4					
X	0.3677	0.3558	0.3544	0.3656	0.3677
Y	0.3603	0.351	0.3416	0.35	0.3603

FE5					
X	0.3558	0.3442	0.3433	0.3544	0.3558
Y	0.351	0.3415	0.3318	0.3416	0.351

FF1					
X	0.397	0.3875	0.3843	0.3931	0.397
Y	0.3689	0.3641	0.3531	0.3587	0.3689

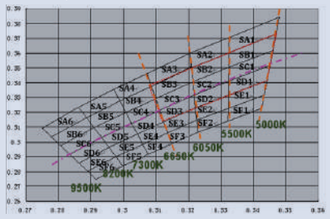
FF2					
X	0.3875	0.3769	0.3742	0.3843	0.3875
Y	0.3641	0.3579	0.3468	0.3531	0.3641

FF3					
X	0.3769	0.3656	0.3636	0.3742	0.3769
Y	0.3579	0.35	0.3388	0.3468	0.3579

FF4					
X	0.3656	0.3544	0.3531	0.3636	0.3656
Y	0.35	0.3416	0.3308	0.3388	0.35

FF5					
X	0.3544	0.3433	0.3425	0.3531	0.3544
Y	0.3416	0.3318	0.3225	0.3308	0.3416

Color Temperature & SW



SA1					
X	0.3479	0.3325	0.3324	0.3469	0.3479
Y	0.3841	0.3706	0.3604	0.3728	0.3841

SA2					
X	0.3325	0.3204	0.3208	0.3324	0.3325
Y	0.3706	0.3597	0.3501	0.3604	0.3706

SA3					
X	0.3204	0.3073	0.3087	0.3208	0.3204
Y	0.3597	0.3463	0.3377	0.3501	0.3597

SA4					
X	0.3073	0.297	0.2989	0.3087	0.3073
Y	0.3463	0.3354	0.3276	0.3377	0.3463

SA5					
X	0.297	0.2862	0.2887	0.2989	0.297
Y	0.3354	0.3235	0.3165	0.3276	0.3354

SA6					
X	0.2862	0.2751	0.2779	0.2887	0.2862
Y	0.3235	0.3093	0.3034	0.3165	0.3235

SB1					
X	0.3469	0.3324	0.3324	0.346	0.3469
Y	0.3728	0.3604	0.3505	0.362	0.3728

SB2					
X	0.3324	0.3208	0.3213	0.3324	0.3324
Y	0.3604	0.3501	0.3408	0.3505	0.3604

SB3					
X	0.3208	0.3087	0.31	0.3213	0.3208
Y	0.351	0.3377	0.3294	0.3408	0.351

SB4					
X	0.3087	0.2989	0.3008	0.31	0.3087
Y	0.3377	0.3276	0.32	0.3294	0.3377

SB5					
X	0.2989	0.2887	0.291	0.3008	0.2989
Y	0.3276	0.3165	0.3096	0.32	0.3276

SB6					
X	0.2887	0.2779	0.2807	0.291	0.2887
Y	0.3165	0.3034	0.2976	0.3096	0.3165

SC1					
X	0.346	0.3324	0.3323	0.345	0.346
Y	0.362	0.3505	0.3409	0.3516	0.362

SC2					
X	0.3324	0.3213	0.3217	0.3323	0.3324
Y	0.3505	0.3408	0.3318	0.3409	0.3505

SC3					
X	0.3213	0.31	0.3112	0.3217	0.3213
Y	0.3408	0.3294	0.3214	0.3318	0.3408

SC4					
X	0.31	0.3008	0.3025	0.3112	0.31
Y	0.3294	0.32	0.3126	0.3214	0.3294

SC5					
X	0.3008	0.291	0.2933	0.3025	0.3008
Y	0.32	0.3096	0.3029	0.3126	0.32

SC6					
X	0.291	0.2807	0.2835	0.2933	0.291
Y	0.3096	0.2976	0.2917	0.3029	0.3096

SD1					
X	0.345	0.3323	0.3322	0.3442	0.345
Y	0.3516	0.3409	0.3317	0.3415	0.3516

SD2					
X	0.3323	0.3217	0.3221	0.3322	0.3323
Y	0.3409	0.3318	0.3231	0.3317	0.3409

SD3					
X	0.3217	0.3112	0.3124	0.3221	0.3217
Y	0.3318	0.3214	0.3136	0.3231	0.3318

SD4					
X	0.3112	0.3025	0.3043	0.3124	0.3112
Y	0.3214	0.3126	0.3054	0.3136	0.3214

SD5					
X	0.3025	0.2933	0.2955	0.3043	0.3025
Y	0.3126	0.3029	0.2965	0.3054	0.3126

SD6					
X	0.2933	0.2835	0.2863	0.2955	0.2933
Y	0.3029	0.2917	0.286	0.2965	0.3029

SE1					
X	0.3442	0.3322	0.3322	0.3433	0.3442
Y	0.3415	0.3317	0.3228	0.3318	0.3415

SE2					
X	0.3322	0.3221	0.3225	0.3322	0.3322
Y	0.3317	0.3231	0.3147	0.3228	0.3317

SE3					
X	0.3221	0.3124	0.3136	0.3225	0.3221
Y	0.3231	0.3136	0.3061	0.3147	0.3231

SE4					
X	0.3124	0.3043	0.3059	0.3136	0.3124
Y	0.3136	0.3054	0.2985	0.3061	0.3136

SE5					
X	0.3043	0.2955	0.2977	0.3059	0.3043
Y	0.3054	0.2965	0.2901	0.2985	0.3054

SE6					
X	0.2955	0.2863	0.2891	0.2977	0.2955
Y	0.2965	0.286	0.2802	0.2901	0.2965

SF1					
X	0.3433	0.3322	0.3321	0.3425	0.3433
Y	0.3318	0.3228	0.3141	0.3225	0.3318

SF2					
X	0.3322	0.3225	0.3229	0.3321	0.3322
Y	0.3228	0.3147	0.3066	0.3141	0.3228

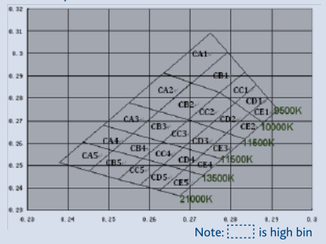
SF3					
X	0.3225	0.3136	0.3148	0.3229	0.3225
Y	0.3147	0.3061	0.2987	0.3066	0.3147

SF4					
X	0.3136	0.3059	0.3076	0.3148	0.3136
Y	0.3061	0.2985	0.2917	0.2987	0.3061

SF5					
X	0.3059	0.2977	0.2998	0.3076	0.3059
Y	0.2985	0.2901	0.284	0.2917	0.2985

SF6					
X	0.2977	0.2891	0.2916	0.2998	0.2977
Y	0.2901	0.2802	0.2749	0.284	0.2901

Color Temperature & CW



Note: is high bin

CA1					
X	0.2751	0.2792	0.2704	0.2635	0.2751
Y	0.3093	0.3007	0.2869	0.2912	0.3093

CA2					
X	0.2635	0.2702	0.2623	0.255	0.2635
Y	0.2915	0.2869	0.2742	0.2779	0.2915

CA3					
X	0.255	0.2623	0.2559	0.2486	0.255
Y	0.2779	0.2742	0.2642	0.2679	0.2779

CA4					
X	0.2486	0.2559	0.2511	0.2438	0.2486
Y	0.2679	0.2642	0.2568	0.2605	0.2679

CA5					
X	0.2379	0.2452	0.2511	0.2438	0.2379
Y	0.2512	0.2475	0.2568	0.2605	0.2512

CB1					
X	0.2792	0.2835	0.2773	0.2704	0.2792
Y	0.3007	0.2917	0.2826	0.2869	0.3007

CB2					
X	0.2704	0.2773	0.2696	0.2623	0.2704
Y	0.2869	0.2826	0.2705	0.2742	0.2869

CB3					
X	0.2623	0.2696	0.2632	0.2559	0.2623
Y	0.2742	0.2705	0.2605	0.2642	0.2742

CB4					
X	0.2559	0.2632	0.2584	0.2511	0.2559
Y	0.2642	0.2605	0.2531	0.2568	0.2642

CB5					
X	0.2511	0.2584	0.2525	0.2452	0.2511
Y	0.2568	0.2531	0.2438	0.2475	0.2568

CC1					
X	0.2835	0.2863	0.2811	0.2773	0.2835
Y	0.2917	0.286	0.2779	0.2826	0.2917

CC2					
X	0.2773	0.2811	0.2744	0.2696	0.2773
Y	0.2826	0.2779	0.2675	0.2705	0.2826

CC3					
X	0.2696	0.2744	0.2683	0.2632	0.2696
Y	0.2705	0.2675	0.2579	0.2605	0.2705

CC4					
X	0.2632	0.2683	0.2635	0.2584	0.2632
Y	0.2605	0.2579	0.2504	0.2531	0.2605

CC5					
X	0.2584	0.2635	0.2577	0.2525	0.2584
Y	0.2531	0.2504	0.2413	0.2438	0.2531

CD1					
X	0.2863	0.2891	0.2846	0.2811	0.2863
Y	0.286	0.2802	0.2733	0.2779	0.286

CD2					
X	0.2811	0.2846	0.279	0.2744	0.2811
Y	0.2779	0.2733	0.2646	0.2675	0.2779

CD3					
X	0.2744	0.279	0.2732	0.2683	0.2744
Y	0.2675	0.2646	0.2554	0.2579	0.2675

CD4					
X	0.2683	0.2732	0.2684	0.2635	0.2683
Y	0.2579	0.2554	0.248	0.2504	0.2579

CD5					
X	0.2635	0.2684	0.2626	0.2577	0.2635
Y	0.2504	0.248	0.2388	0.2413	0.2504

CE1					
X	0.2891	0.2916	0.288	0.2846	0.2891
Y	0.2802	0.2749	0.2691	0.2733	0.2802

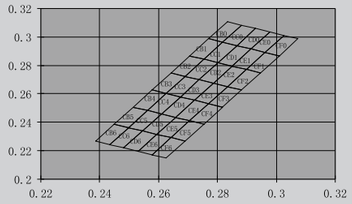
CE2					
X	0.2846	0.288	0.2834	0.279	0.2846
Y	0.2733	0.2691	0.2618	0.2646	0.2733

CE3					
X	0.279	0.2834	0.2779	0.2732	0.279
Y	0.2646	0.2618	0.253	0.2554	0.2646

CE4					
X	0.2732	0.2779	0.2732	0.2684	0.2732
Y	0.2554	0.253	0.2455	0.248	0.2554

CE5					
X	0.2684	0.2732	0.2675	0.2626	0.2684
Y	0.248	0.2455	0.2364	0.2388	0.248

Chromaticity Bin

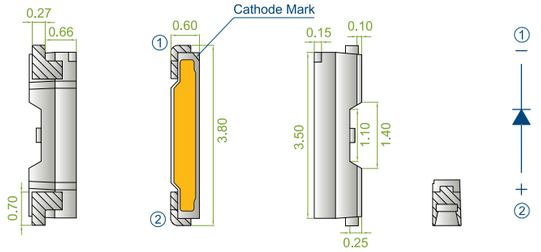


CB0					
X	0.2834	0.2882	0.2818	0.277	0.2834
Y	0.3107	0.3083	0.2963	0.2987	0.3107
CB1					
X	0.277	0.2818	0.2754	0.2706	0.277
Y	0.2987	0.2963	0.2843	0.2867	0.2987
CB2					
X	0.2706	0.2754	0.269	0.2642	0.2706
Y	0.2867	0.2843	0.2723	0.2747	0.2867
CB3					
X	0.2642	0.269	0.2626	0.2578	0.2642
Y	0.2747	0.2723	0.2603	0.2627	0.2747
CB4					
X	0.2578	0.2626	0.2562	0.2514	0.2578
Y	0.2627	0.2603	0.2483	0.2507	0.2627
CB5					
X	0.2514	0.2562	0.2498	0.245	0.2514
Y	0.2507	0.2483	0.2363	0.2387	0.2507
CB6					
X	0.245	0.2498	0.2434	0.2386	0.245
Y	0.2387	0.2363	0.2243	0.2267	0.2387
CC0					
X	0.2882	0.293	0.2866	0.2818	0.2882
Y	0.3083	0.3059	0.2939	0.2963	0.3083
CC1					
X	0.2818	0.2866	0.2802	0.2754	0.2818
Y	0.2963	0.2939	0.2819	0.2843	0.2963
CC2					
X	0.2754	0.2802	0.2738	0.269	0.2754
Y	0.2843	0.2819	0.2699	0.2723	0.2843
CC3					
X	0.269	0.2738	0.2674	0.2626	0.269
Y	0.2723	0.2699	0.2579	0.2603	0.2723
CC4					
X	0.2626	0.2674	0.261	0.2562	0.2626
Y	0.2603	0.2579	0.2459	0.2483	0.2603
CC5					
X	0.2562	0.261	0.2546	0.2498	0.2562
Y	0.2483	0.2459	0.2339	0.2363	0.2483
CC6					
X	0.2498	0.2546	0.2482	0.2434	0.2498
Y	0.2363	0.2339	0.2219	0.2243	0.2363
CD0					
X	0.293	0.2978	0.2914	0.2866	0.293
Y	0.3059	0.3035	0.2915	0.2939	0.3059
CD1					
X	0.2866	0.2914	0.285	0.2802	0.2866
Y	0.2939	0.2915	0.2795	0.2819	0.2939
CD2					
X	0.2802	0.285	0.2786	0.2738	0.2802
Y	0.2819	0.2795	0.2675	0.2699	0.2819

CD3					
X	0.2738	0.2786	0.2722	0.2674	0.2738
Y	0.2699	0.2675	0.2555	0.2579	0.2699
CD4					
X	0.2674	0.2722	0.2658	0.261	0.2674
Y	0.2579	0.2555	0.2435	0.2459	0.2579
CD5					
X	0.261	0.2658	0.2594	0.2546	0.261
Y	0.2459	0.2435	0.2315	0.2339	0.2459
CD6					
X	0.2546	0.2594	0.253	0.2482	0.2546
Y	0.2339	0.2315	0.2195	0.2219	0.2339
CE0					
X	0.2978	0.3026	0.2962	0.2914	0.2978
Y	0.3035	0.3011	0.2891	0.2915	0.3035
CE1					
X	0.2914	0.2962	0.2898	0.285	0.2914
Y	0.2915	0.2891	0.2771	0.2795	0.2915
CE2					
X	0.285	0.2898	0.2834	0.2786	0.285
Y	0.2795	0.2771	0.2651	0.2675	0.2795
CE3					
X	0.2786	0.2834	0.277	0.2722	0.2786
Y	0.2675	0.2651	0.2531	0.2555	0.2675
CE4					
X	0.2722	0.277	0.2706	0.2658	0.2722
Y	0.2555	0.2531	0.2411	0.2435	0.2555
CE5					
X	0.2658	0.2706	0.2642	0.2594	0.2658
Y	0.2435	0.2411	0.2291	0.2315	0.2435
CE6					
X	0.2594	0.2642	0.2578	0.253	0.2594
Y	0.2315	0.2291	0.2171	0.2195	0.2315
CF0					
X	0.3026	0.3074	0.301	0.2962	0.3026
Y	0.3011	0.2987	0.2867	0.2891	0.3011
CF1					
X	0.2962	0.301	0.2946	0.2898	0.2962
Y	0.2891	0.2867	0.2747	0.2771	0.2891
CF2					
X	0.2898	0.2946	0.2882	0.2834	0.2898
Y	0.2771	0.2747	0.2627	0.2651	0.2771
CF3					
X	0.2834	0.2882	0.2818	0.277	0.2834
Y	0.2651	0.2627	0.2507	0.2531	0.2651
CF4					
X	0.277	0.2818	0.2754	0.2706	0.277
Y	0.2531	0.2507	0.2387	0.2411	0.2531
CF5					
X	0.2706	0.2754	0.269	0.2642	0.2706
Y	0.2411	0.2387	0.2267	0.2291	0.2411
CF6					
X	0.2642	0.269	0.2626	0.2578	0.2642
Y	0.2291	0.2267	0.2147	0.2171	0.2291

TO-3806

RoHS Pb Free



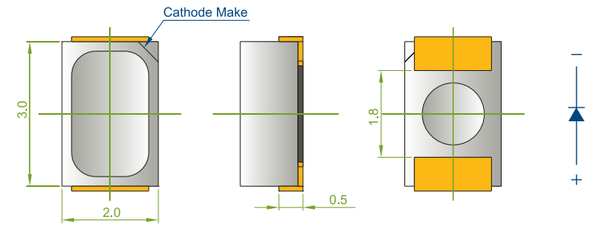
Features
- 3.8mm x 0.6mm x 1.0mm
- High efficiency LED
- IC compatible
- Compatible with automatic placement equipment
- Compatible with infrared and vapor phase reflow solder process

Application
- Backlight

Part Number	Wavelength λ_d(nm) Cx/Cy	Emitting Color	Forward Voltage (V)	Test Condition (mA)	IV(mcd)	Viewing Angle 2θ½ (°)
TO-3806CWLQ-D1	0.295/0.285	White	2.8-3.4	20	2400-2700	120
TO-3806CWLQ-D1	0.28/0.26				2200-2500	
TO-3806CWLQ-D1	0.263/0.235				2300-2000	
TO-3806CWLQ-E2	0.295/0.285				2500-2800	
TO-3806CWLQ-E2	0.28/0.26				2300-2600	
TO-3806CWLQ-E2	0.263/0.235				2100-2400	
TO-3806CWLQ-F4	0.295/0.285				2700-3000	
TO-3806CWLQ-F4	0.28/0.26				2500-2800	
TO-3806CWLQ-F4	0.263/0.235				2300-2600	

TO-3020

RoHS Pb Free



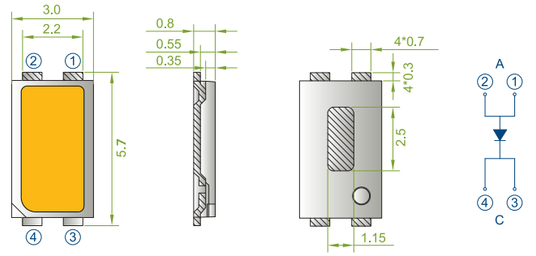
Features
- 3.0mm x 2.0mm x 2.0mm
- High efficiency LED
- IC compatible
- Compatible with automatic placement equipment
- Compatible with infrared and vapor phase reflow solder process

Application
- Backlight

Part Number	Wavelength λ_d(nm) Cx/Cy	Emitting Color	Forward Voltage (V)	Test Condition (mA)	IV(mcd)	Viewing Angle 2θ½ (°)
TO-3020CWLQ-D1	0.295/0.285	White	2.8-3.4	20	2400-2800	120
TO-3020CWLQ-D1	0.28/0.26				2200-2600	
TO-3020CWLQ-D1	0.263/0.235				2000-2400	
TO-3020CWLQ-E2	0.295/0.285				2500-2900	
TO-3020CWLQ-E2	0.28/0.26				2300-2700	
TO-3020CWLQ-E2	0.263/0.235				2100-2500	
TO-3020CWLQ-F4	0.295/0.285				2700-3100	
TO-3020CWLQ-F4	0.28/0.26				2500-2900	
TO-3020CWLQ-F4	0.263/0.235				2300-2700	

SMD

TO-5730

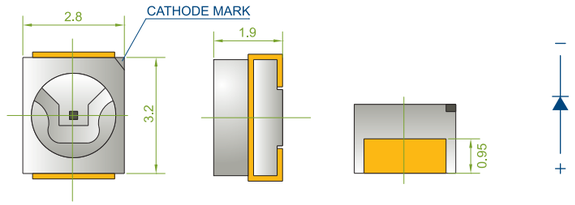


Features
- 5.7mm x 3.0mm x 0.8mm
- High efficiency LED
- IC compatible
- Compatible with automatic placement equipment
- Compatible with infrared and vapor phase reflow solder process

Application
- General lighting
- Backlight

Part Number	Wavelength λ_d(nm) Cx/Cy	Emitting Color	Forward Voltage (V)	Test Condition (mA)	IV(mcd)	Viewing Angle 2θ½ (°)
TO-5730CWLQ-H1	0.295/0.285	White	2.8-3.4	60	8750-10850	120
TO-5730CWLQ-H1	0.28/0.26			60	7000-9100	
TO-5730CWLQ-H1	0.263/0.235			60	5250-7350	
TO-5730CWLQ-H2	0.295/0.285			120	15050-17150	
TO-5730CWLQ-H2	0.28/0.26			120	13300-15400	
TO-5730CWLQ-H2	0.263/0.235			120	11550-13650	
TO-5730CWLQ-H3	0.295/0.285			150	15750-17850	
TO-5730CWLQ-H3	0.28/0.26			150	14000-16100	
TO-5730CWLQ-H3	0.263/0.235			150	12250-14350	

TO-3228
TO-3528



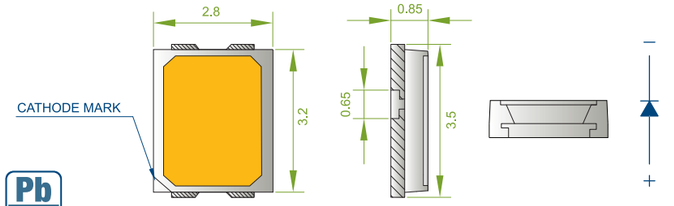
Features
- 3.5mm x 2.8mm x 1.9mm
- High efficiency LED
- IC compatible
- Compatible with automatic placement equipment
- Compatible with infrared and vapor phase reflow solder process

Application
- General lighting

Part Number	Wavelength λ_d(nm) Cx/Cy	Emitting Color	Color Temperature	Forward Voltage (V)	Test Condition (mA)	Typ Luminous Flux(LM) @20mA	CRI (Min)	Viewing Angle 2θ½ (°)
3528 0.06W/20mA								
TO-3228TWLQ-E2	0.43/0.40	Warm White	2600K-3500K	2.8-3.4	20	6.5	70	120
TO-3228FWLQ-E2	0.37/0.37	Neutral White	3500K-5000K			7.5	70	
TO-3228SWLQ-E2	0.32/0.33	Cool White	5000K-9500K			7.5	70	
TO-3228TWNQ-E2	0.43/0.40	Warm White	2600K-3500K			6	80	
TO-3228FWNQ-E2	0.37/0.37	Neutral White	3500K-5000K			7	80	
TO-3228SWNQ-E2	0.32/0.33	Cool White	5000K-9500K			7	80	
TO-3228TWLQ-F2	0.43/0.40	Warm White	2600K-3500K			7.5	70	
TO-3228FWLQ-F2	0.37/0.37	Neutral White	3500K-5000K			8.5	70	
TO-3228SWLQ-F2	0.32/0.33	Cool White	5000K-9500K			8.5	70	
TO-3228TWNQ-F2	0.43/0.40	Warm White	2600K-3500K			7	80	
TO-3228FWNQ-F2	0.37/0.37	Neutral White	3500K-5000K			8	80	
TO-3228SWNQ-F2	0.32/0.33	Cool White	5000K-9500K			8	80	

SMD

TO-2835

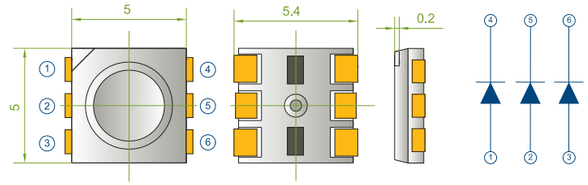


Features
- 3.5mm x 2.8mm x 0.85mm
- High efficiency LED
- IC compatible
- Compatible with automatic placement equipment
- Compatible with infrared and vapor phase reflow solder process

Application
- General lighting

Part Number	Wavelength λ_d(nm) Cx/Cy	Emitting Color	Color Temperature	Forward Voltage (V)	Test Condition (mA)	Typ Luminous Flux(LM)	CRI (Min)	Viewing Angle 2θ½ (°)
2835 0.2W/60mA						@60mA		
TO-2835TWLQ-H1	0.43/0.40	Warm White	2600K-3500K	2.8-3.4	60	25	70	120
TO-2835FWLQ-H1	0.37/0.37	Neutral White	3500K-5000K			27	70	
TO-2835SWLQ-H1	0.32/0.33	Cool White	5000K-9500K			27	70	
TO-2835TWNQ-H1	0.43/0.40	Warm White	2600K-3500K			24	80	
TO-2835FWNQ-H1	0.37/0.37	Neutral White	3500K-5000K			26	80	
TO-2835SWNQ-H1	0.32/0.33	Cool White	5000K-9500K			26	80	
2835 0.4W/120mA						@120mA		
TO-2835TWLQ-H2	0.43/0.40	Warm White	2600K-3500K	2.8-3.4	120	45	70	120
TO-2835FWLQ-H2	0.37/0.37	Neutral White	3500K-5000K			47	70	
TO-2835SWLQ-H2	0.32/0.33	Cool White	5000K-9500K			47	70	
TO-2835TWNQ-H2	0.43/0.40	Warm White	2600K-3500K			42	80	
TO-2835FWNQ-H2	0.37/0.37	Neutral White	3500K-5000K			44	80	
TO-2835SWNQ-H2	0.32/0.33	Cool White	5000K-9500K			44	80	
2835 0.5W/150mA						@150mA		
TO-2835TWLQ-H3	0.43/0.40	Warm White	2600K-3500K	2.8-3.4	150	50	70	120
TO-2835FWLQ-H3	0.37/0.37	Neutral White	3500K-5000K			53	70	
TO-2835SWLQ-H3	0.32/0.33	Cool White	5000K-9500K			53	70	
TO-2835TWNQ-H3	0.43/0.40	Warm White	2600K-3500K			46	80	
TO-2835FWNQ-H3	0.37/0.37	Neutral White	3500K-5000K			50	80	
TO-2835SWNQ-H3	0.32/0.33	Cool White	5000K-9500K			50	80	

TO-5050



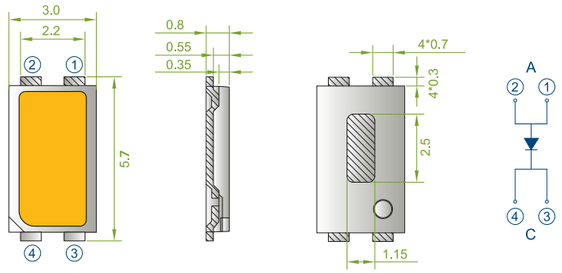
RoHS Pb Free

Features
- 5.0mm x 5.0mm x 1.55mm
- High efficiency LED
- IC compatible
- Compatible with automatic placement equipment
- Compatible with infrared and vapor phase reflow solder process

Application
- General lighting

Part Number	Wavelength λ_d(nm) Cx/Cy	Emitting Color	Color Temperature	Forward Voltage (V)	Test Condition (mA)	Typ Luminous Flux(LM) @60mA	CRI (Min)	Viewing Angle 2θ½ (°)
5050 0.2W/60mA								
TO-5050TWLQ-E2	0.43/0.40	Warm White	2600K-3500K	2.8-3.4	60	22	70	120
TO-5050FWLQ-E2	0.37/0.37	Neutral White	3500K-5000K			24	70	
TO-5050SWLQ-E2	0.32/0.33	Cool White	5000K-9500K			24	70	
TO-5050TWNQ-E2	0.43/0.40	Warm White	2600K-3500K			20	80	
TO-5050FWNQ-E2	0.37/0.37	Neutral White	3500K-5000K			22	80	
TO-5050SWNQ-E2	0.32/0.33	Cool White	5000K-9500K			22	80	
TO-5050TWLQ-F2	0.43/0.40	Warm White	2600K-3500K			24	70	
TO-5050FWLQ-F2	0.37/0.37	Neutral White	3500K-5000K			26	70	
TO-5050SWLQ-F2	0.32/0.33	Cool White	5000K-9500K			26	70	
TO-5050TWNQ-F2	0.43/0.40	Warm White	2600K-3500K			22	80	
TO-5050FWNQ-F2	0.37/0.37	Neutral White	3500K-5000K			24	80	
TO-5050SWNQ-F2	0.32/0.33	Cool White	5000K-9500K			24	80	

TO-5730



RoHS Pb Free

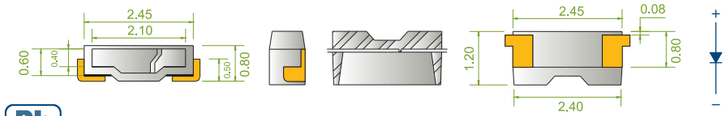
Features
- 5.7mm x 3.0mm x 0.8mm
- High efficiency LED
- IC compatible
- Compatible with automatic placement equipment
- Compatible with infrared and vapor phase reflow solder process

Application
- General lighting
- Backlight

Part Number	Wavelength λ_d(nm) Cx/Cy	Emitting Color	Color Temperature	Forward Voltage (V)	Test Condition (mA)	Typ Luminous Flux(LM)	CRI (Min)	Viewing Angle 2θ½ (°)
5730 0.2W/60mA					@60mA			
TO-5730TWLQ-H1	0.43/0.40	Warm White	2600K-3500K	2.8-3.4	60	25	70	120
TO-5730FWLQ-H1	0.37/0.37	Neutral White	3500K-5000K			27	70	
TO-5730SWLQ-H1	0.32/0.33	Cool White	5000K-9500K			27	70	
TO-5730TWNQ-H1	0.43/0.40	Warm White	2600K-3500K			24	80	
TO-5730FWNQ-H1	0.37/0.37	Neutral White	3500K-5000K			26	80	
TO-5730SWNQ-H1	0.32/0.33	Cool White	5000K-9500K			26	80	
5730 0.4W/120mA					@120mA			
TO-5730TWLQ-H2	0.43/0.40	Warm White	2600K-3500K	2.8-3.4	120	46	70	120
TO-5730FWLQ-H2	0.37/0.37	Neutral White	3500K-5000K			48	70	
TO-5730SWLQ-H2	0.32/0.33	Cool White	5000K-9500K			48	70	
TO-5730TWNQ-H2	0.43/0.40	Warm White	2600K-3500K			43	80	
TO-5730FWNQ-H2	0.37/0.37	Neutral White	3500K-5000K			45	80	
TO-5730SWNQ-H2	0.32/0.33	Cool White	5000K-9500K			45	80	
5730 0.5W/150mA					@150mA			
TO-5730TWLQ-H3	0.43/0.40	Warm White	2600K-3500K	2.8-3.6	150	52	70	120
TO-5730FWLQ-H3	0.37/0.37	Neutral White	3500K-5000K			55	70	
TO-5730SWLQ-H3	0.32/0.33	Cool White	5000K-9500K			55	70	
TO-5730TWNQ-H3	0.43/0.40	Warm White	2600K-3500K			48	80	
TO-5730FWNQ-H3	0.37/0.37	Neutral White	3500K-5000K			52	80	
TO-5730SWNQ-H3	0.32/0.33	Cool White	5000K-9500K			52	80	

TO-2808

PLCC 2808 Package



RoHS Pb Free

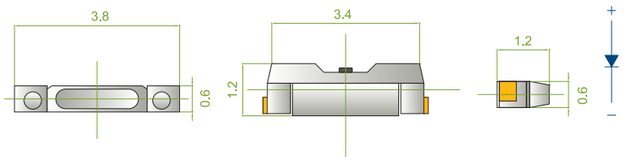
Features
- 2.8mm x 0.8mm x 1.2mm
- High efficiency LED
- Low power consumption
- Wide viewing angle
- Various color available

Application
- LCD backlight
- Indicator light

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-2808SC-BF	InGaN	Blue	468	Water Clear	3.2	3.6	146	281	120
TO-2808SC-PF	InGaN	Pure Green	525		3.2	3.6	146	281	
TO-2808SC-MGE	AlGaInP	High Bright Green	572		2.0	2.4	51	98	
TO-2808SC-MYF	AlGaInP	High Bright Yellow	590		2.0	2.4	146	281	
TO-2808SC-MRE	AlGaInP	High Bright Red	630		1.9	2.4	51	98	

TO-3806

PLCC 3806 Package



RoHS Pb Free

Features
- 3.8mm x 0.6mm x 1.2mm
- High efficiency LED
- Low power consumption
- Wide viewing angle
- Various color available

Application
- LCD backlight
- Indicator light

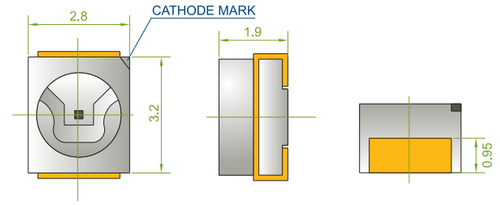
Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA	Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	TYP.	
TO-3806SC-BF	InGaN	Blue	468	Water Clear	3.2	3.6	200	120
TO-3806SC-PG	InGaN	Pure Green	525		3.2	3.6	450	
TO-3806SC-MGE	AlGaInP	High Bright Green	572		2.0	2.4	60	
TO-3806SC-MAF	AlGaInP	High Bright Amber	605		2.0	2.4	155	
TO-3806SC-MRF	AlGaInP	High Bright Red	630		1.9	2.4	150	

TO-3228
TO-3528

PLCC 3228 Package

Single Color

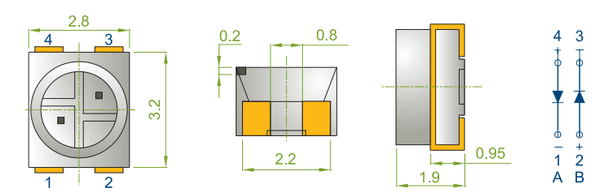
RoHS Pb Free



Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-3228BC-BF	InGaN	Blue	468	Water Clear	3.3	3.6	146	281	120
TO-3228BC-PG	InGaN	Pure Green	525		3.3	3.6	417	805	
TO-3228BC-MGE	AlGaInP	High Bright Green	572		2.2	2.4	51	98	
TO-3228BC-MYG	AlGaInP	High Bright Yellow	592		2.0	2.4	417	805	
TO-3228BC-MAF	AlGaInP	High Bright Amber	605		2.0	2.4	542	1046	
TO-3228BC-ED	GaAsP	Orange	621		2.0	2.4	10.5	20	
TO-3228BC-MEG	AlGaInP	High Bright Orange	623		2.0	2.4	417	805	
TO-3228BC-MRG	AlGaInP	High Bright Red	630		1.9	2.4	417	667	

Double Color

RoHS Pb Free

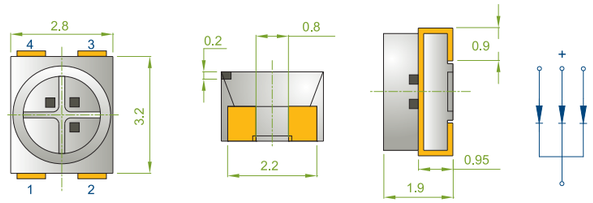


Features
- 3.2mm x 2.8mm x 1.9mm
- High efficiency LED
- Low power consumption
- Wide viewing angle
- Dual color
- Various color available

Application
- Backlight
- Sign display
- General lighting
- Wall wash lighting
- Residential lighting

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-3228BC-PBFF	InGaN	Pure Green	525	Water Clear	3.3	3.6	51	98	120
	InGaN	Blue	468		3.3	3.6	51	98	
TO-3228BC-MGMGEE	AlGaInP	High Bright Green	572		2.0	2.4	51	98	
	AlGaInP	High Bright Green	572		2.0	2.4	51	98	
TO-3228BC-MYPFF	AlGaInP	High Bright Yellow	590		2.0	2.4	146	281	
	InGaN	Pure Green	525		3.3	3.6	146	281	
TO-3228BC-MYMGFE	AlGaInP	High Bright Yellow	590		2.0	2.4	146	281	
	AlGaInP	High Bright Green	572		2.0	2.4	51	98	
TO-3228BC-MAMAFF	AlGaInP	High Bright Amber	605		2.0	2.4	146	281	
	AlGaInP	High Bright Amber	605		2.0	2.4	146	281	
TO-3228BC-MEPFG	AlGaInP	High Bright Orange	623		2.0	2.4	51	98	
	InGaN	Pure Green	525		2.0	2.4	146	281	
TO-3228BC-MEMYFF	AlGaInP	High Bright Orange	623		2.0	2.4	146	281	
	AlGaInP	High Bright Yellow	590		2.0	2.4	146	281	
TO-3228BC-MRBEF	AlGaInP	High Bright Red	630		1.9	2.4	51	98	
	InGaN	Blue	468		3.3	3.6	146	281	
TO-3228BC-MRPFG	AlGaInP	High Bright Red	630		1.9	2.4	146	281	
	InGaN	Pure Green	525		3.3	3.6	417	805	
TO-3228BC-MRMGEE	AlGaInP	High Bright Red	630		1.9	2.4	51	98	
	AlGaInP	High Bright Green	572		2.0	2.4	51	98	

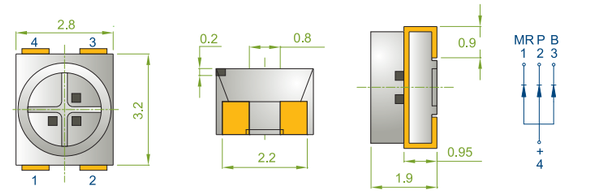
Full Color



Features
- 3.2mm x 2.8mm x 1.9mm
- High efficiency LED
- Low power consumption
- Wide viewing angle

Application
- Backlight
- Sign display
- General lighting
- Wall wash lighting
- Residential lighting

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-3228BC-3MYH	AlGaInP	High Bright Yellow	590	Water Clear	2.0	2.4	917	1997	
TO-3228BC-3MAK	AlGaInP	High Bright Amber	605		2.0	2.4	1150	2278	



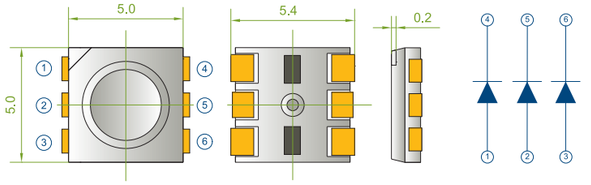
Features
- 3.2mm x 2.8mm x 1.9mm
- High efficiency LED
- Low power consumption
- Wide viewing angle
- RGB full color

Application
- Backlight
- Sign display
- General lighting
- Wall wash lighting
- Residential lighting

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-3228BC-MRPBFGF	AlGaInP	High Bright Red	630	Water Clear	2.0	2.4	146	281	120
	InGaN	Pure Green	525		3.2	3.6	417	667	
	InGaN	Blue	468		3.2	3.6	146	281	
TO-3228BC-MEPBFGF	AlGaInP	High Bright Orange	623		2.0	2.4	146	281	
	InGaN	Pure Green	525		3.2	3.6	417	667	
	InGaN	Blue	468		3.2	3.6	146	281	

TO-5050

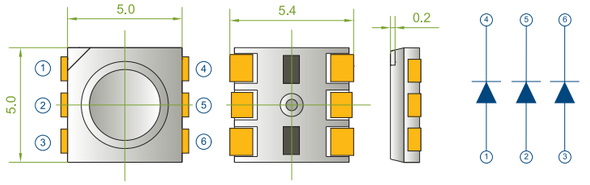
PLCC 5050 Package



Features
- 5.0mm x 5.0mm x 1.5mm
- High efficiency LED
- Low power consumption
- Wide viewing angle
- Various color available

Application
- Office lighting
- Indoor display
- Residential lighting
- General lighting
- Backlight and indicator

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-5050BC-143PK	InGaN	Pure Green	525	Water Clear	3.2	3.6	2015	2711	120
TO-5050BC-123MYH	AlGaInP	High Bright Yellow	590		2.0	2.4	917	1466	
TO-5050BC-3MAG	AlGaInP	High Bright Amber	605		2.0	2.4	417	805	
TO-5050BC-143MEH-N	AlGaInP	High Bright Orange	623		2.0	2.4	917	1466	
TO-5050BC-123MRG	AlGaInP	High Bright Red	630		1.9	2.4	419	667	



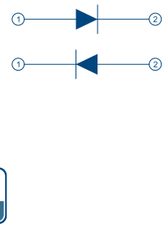
Features
- 5.0mm x 5.0mm x 1.5mm
- High efficiency LED
- Low power consumption
- Wide viewing angle
- RGB full color

Application
- Office lighting
- Indoor display
- Residential lighting
- General lighting
- Backlight and indicator

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-5050BC-MEPBGHF	AlGaInP	High Bright Orange	623	Water Clear	2.0	2.4	417	667	120
	InGaN	Pure Green	525		3.2	3.6	417	667	
	InGaN	Blue	468		3.2	3.6	146	281	
TO-5050BC-MRPBFGF	AlGaInP	High Bright Red	630		1.9	2.4	146	281	
	InGaN	Pure Green	525		3.2	3.6	417	667	
	InGaN	Blue	468		3.2	3.6	146	281	

TO-1005

PCB 1005 Package



RoHS Pb Free

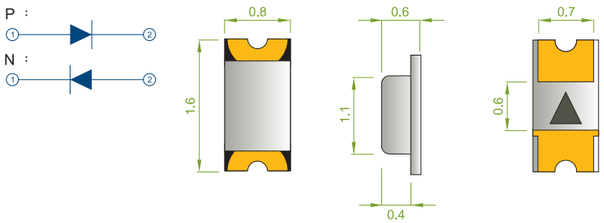
Features
- 1.0mm x 0.5mm x 0.43mm
- Low power consumption
- Wide viewing angle
- Various color available

Application
- Indicator light
- Backlight
- Automotive interior light
- Indoor display board

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-1005BC-BE	InGaN	Blue	468	Water Clear	3.2	3.6	51	112	120
TO-1005BC-PF	InGaN	Pure Green	525		3.2	3.6	146	247	
TO-1005BC-MGE	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-1005BC-MYF	AlGaInP	High Bright Yellow	592		2.0	2.4	146	247	
TO-1005BC-MAE	AlGaInP	High Bright Amber	605		2.0	2.4	51	112	
TO-1005BC-MEE	AlGaInP	High Bright Orange	623		2.0	2.4	51	112	
TO-1005BC-MRE	AlGaInP	High Bright Red	630		2.0	2.4	51	112	

TO-1608

PCB 1608 Package



RoHS Pb Free

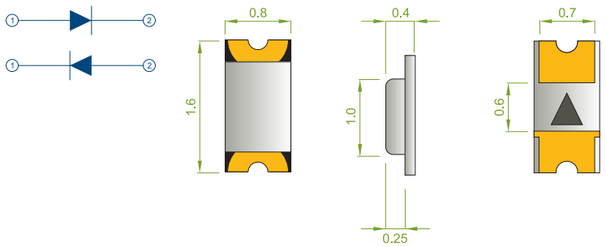
Features
- 1.6mm x 0.8mm x 0.6mm
- Low power consumption
- Wide viewing angle
- Various color available

Application
- Indicator light
- Backlight
- Automotive interior light
- Indoor display board

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-1608BC-BF	InGaN	Blue	468	Water Clear	3.2	3.6	112	247	120
TO-1608BC-PG	InGaN	Pure Green	525		3.2	3.6	321	705	
TO-1608BC-MGE	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-1608BC-MYF	AlGaInP	High Bright Yellow	592		2.0	2.4	146	247	
TO-1608BC-MAE	AlGaInP	High Bright Amber	605		2.0	2.4	51	112	
TO-1608BC-MEE	AlGaInP	High Bright Orange	623		2.0	2.4	51	112	
TO-1608BC-MRE	AlGaInP	High Bright Red	630		2.0	2.4	51	112	

TO-A1608

PCB 1608 Package



RoHS Pb Free

Features
- 1.6mm x 0.8mm x 0.4mm
- Low power consumption
- Wide viewing angle
- Various color available

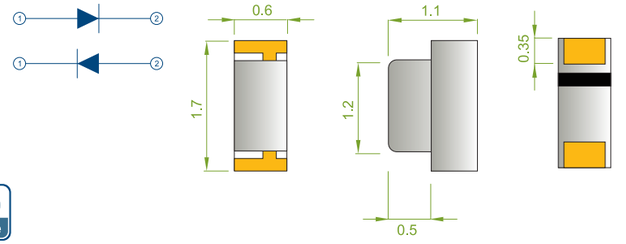
Application
- Indicator light
- Backlight
- Automotive interior light
- Indoor display board

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-A1608BC-BF	InGaN	Blue	468	Water Clear	3.2	3.6	112	247	120
TO-A1608BC-PG	InGaN	Pure Green	525		3.2	3.6	321	705	
TO-A1608BC-MGE	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-A1608BC-MYF	AlGaInP	High Bright Yellow	592		2.0	2.4	146	247	
TO-A1608BC-MAE	AlGaInP	High Bright Amber	605		2.0	2.4	51	112	
TO-A1608BC-MEE	AlGaInP	High Bright Orange	623		2.0	2.4	51	112	
TO-A1608BC-MRE	AlGaInP	High Bright Red	630		2.0	2.4	51	112	

Application
- Indicator light
- Backlight
- Automotive interior light
- Indoor display board

TO-1706

PCB 1706 Package



RoHS Pb Free

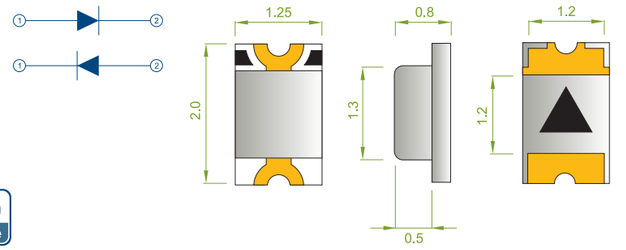
Features
- 1.7mm x 0.6mm x 1.1mm
- Low power consumption
- Wide viewing angle
- Various color available

Application
- Indicator light
- Backlight
- Automotive interior light
- Indoor display board

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-1706BC-BF	InGaN	Blue	468	Water Clear	3.2	3.6	112	247	120
TO-1706BC-PG	InGaN	Pure Green	525		3.2	3.6	321	705	
TO-1706BC-MGE	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-1706BC-MYF	AlGaInP	High Bright Yellow	592		2.0	2.4	146	247	
TO-1706BC-MAF	AlGaInP	High Bright Amber	605		2.0	2.4	146	247	
TO-1706BC-MEF	AlGaInP	High Bright Orange	623		2.0	2.4	146	247	
TO-1706BC-MRE	AlGaInP	High Bright Red	630		2.0	2.4	51	112	

TO-2013

PCB 2013 Package



RoHS Pb Free

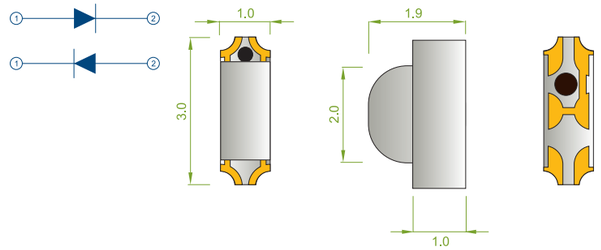
Features
- 2.0mm x 1.25mm x 0.8mm
- Low power consumption
- Wide viewing angle
- Various color available

Application
- Indicator light
- Backlight
- Automotive interior light
- Indoor display board

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-2013BC-BF	InGaN	Blue	468	Water Clear	3.2	3.6	112	247	120
TO-2013BC-PG	InGaN	Pure Green	525		3.2	3.6	321	705	
TO-2013BC-MGE	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-2013BC-MYF	AlGaInP	High Bright Yellow	592		2.0	2.4	146	247	
TO-2013BC-MAF	AlGaInP	High Bright Amber	605		2.0	2.4	146	247	
TO-2013BC-MEF	AlGaInP	High Bright Orange	623		2.0	2.4	146	247	
TO-2013BC-MRE	AlGaInP	High Bright Red	630		2 0	2 4	51	112	

TO-3010

PCB 3010 Package



RoHS Pb Free

Features
- 3.0mm x 1.0mm x 1.9mm
- Low power consumption
- Wide viewing angle
- Various color available

Application
- Indicator light
- Backlight
- Automotive interior light
- Indoor display board

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_v(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-3010AC-BE	InGaN	Blue	468	Water Clear	3.2	3.6	51	112	120
TO-3010AC-PF	InGaN	Pure Green	525		3.2	3.6	146	247	
TO-3010AC-MGE	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-3010AC-MYF	AlGaInP	High Bright Yellow	592		2.0	2.4	146	247	
TO-3010AC-MAE	AlGaInP	High Bright Amber	605		2.0	2.4	51	112	
TO-3010AC-MEE	AlGaInP	Super Bright Orange	623		2.0	2.4	51	112	
TO-3010AC-MRE	AlGaInP	Ultra Bright Red	630		2 0	2 4	51	112	

TO-3216

PCB 3216 Package

RoHS Pb Free

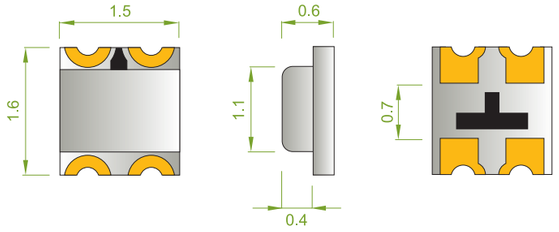
Features
- 3.2mm x 1.6mm x 1.1mm
- Low power consumption
- Wide viewing angle
- Various color available

Application
- Indicator light
- Backlight
- Automotive interior light
- Indoor display board

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_v(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-3216BC-BF	InGaN	Blue	468	Water Clear	3.2	3.6	112	247	120
TO-3216BC-PG	InGaN	Pure Green	525		3.2	3.6	321	705	
TO-3216BC-MGE	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-3216BC-MYF	AlGaInP	High Bright Yellow	592		2.0	2.4	146	247	
TO-3216BC-MAF	AlGaInP	High Bright Amber	605		2.0	2.4	146	247	
TO-3216BC-MEF	AlGaInP	Super Bright Orange	623		2.0	2.4	146	247	
TO-3216BC-MRE	AlGaInP	Ultra Bright Red	630		2 0	2 4	51	112	

TO-1615

PCB 1615 Package



RoHS Pb Free

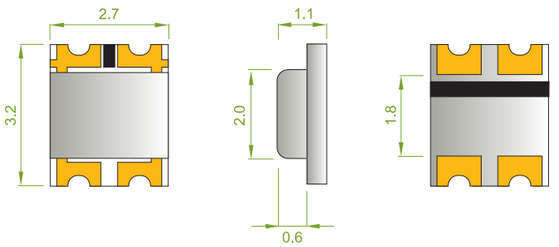
Features
- 1.6mm x 1.5 mm x 0.6 mm
- Dual color
- Low power consumption
- Wide viewing angle
- Various color available

Application
- Indicator light
- Backlight
- Automotive interior light
- Indoor display board

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_v(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-1615BC-MAMGEE	AlGaInP	High Bright Amber	605	Water Clear	2.0	2.4	51	112	120
	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-1615BC-MEBEE	AlGaInP	High Bright Orange	623		2.0	2.4	51	112	
	InGaN	Blue	468		3.2	3.6	51	112	
TO-1615BC-MEPEF	AlGaInP	High Bright Orange	623		2.0	2.4	51	112	
	InGaN	Pure Green	525		3.2	3.6	146	247	
TO-1615BC-MEMGEE	AlGaInP	High Bright Orange	623		2.0	2.4	51	112	
	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-1615BC-MRBFF	AlGaInP	High Bright Red	630		2.0	2.4	146	247	
	InGaN	Blue	468		3.2	3.6	146	247	
TO-1615BC-MRMAEE	AlGaInP	High Bright Red	630		2.0	2.4	51	112	
	AlGaInP	High Bright Amber	605		2.0	2.4	51	112	
TO-1615BC-MRMGEE	AlGaInP	High Bright Red	630		2.0	2.4	51	112	
	AlGaInP	High Bright Green	572		2.0	2.4	51	112	

TO-3227

PCB 3227 Package



RoHS Pb Free

Features
- 3.2mm x 2.7mm x 1.1 mm
- Dual color
- Low power consumption
- Wide viewing angle
- Various color available

Application
- Indicator light
- Backlight
- Automotive interior light
- Indoor display board

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_v(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TO-3227BC-MYBEE	AlGaInP	High Bright Yellow	590	Water Clear	2.0	2.4	51	112	120
	InGaN	Blue	468		3.2	3.6	51	112	
TO-3227BC-MYMGEE	AlGaInP	High Bright Yellow	590		2.0	2.4	51	112	
	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-3227BC-MABFF	AlGaInP	High Bright Amber	605		2.0	2.4	146	247	
	InGaN	Blue	468		3.2	3.6	146	247	
TO-3227BC-MAMGEE	AlGaInP	High Bright Amber	605		2.0	2.4	51	112	
	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-3227BC-MEMGEE	AlGaInP	High Bright Orange	623		2.0	2.4	51	112	
	AlGaInP	High Bright Green	572		2.0	2.4	51	112	
TO-3227BC-MRBEE	AlGaInP	High Bright Red	630		2.0	2.4	51	112	
	InGaN	Blue	468		3.2	3.6	51	112	
TO-3227BC-MRPEF	AlGaInP	High Bright Red	630		2.0	2.4	51	112	
	InGaN	Pure Green	525		3.2	3.6	146	247	
TO-3227BC-MRMGEE	AlGaInP	High Bright Red	630		2.0	2.4	51	112	
	AlGaInP	High Bright Green	572		2.0	2.4	51	112	

HIGH POWER

TOP (1W/3W)

RED (AlGaInP)

Part Number	Forward Voltage (V)			Luminous Flux (Lm)			Dominant wavelength (nm)			Test Condition
	Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
TOP-1RD1	2.0	2.2	2.4	30	40	50	620.0	622.5	625.0	IF=350mA
TOP-3RG1	2.1	2.3	2.5	70	80	90	620.0	622.5	625.0	IF=700mA

Features
- High brightness, high performance
- High color purity, good uniformity
- Low light decline, long working life

Application
- Generial Lighting

TOP (1W/3W)

GREEN (InGaN)

Part Number	Forward Voltage (V)			Luminous Flux (Lm)			Dominant wavelength (nm)			Test Condition
	Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
TOP-1GF2	3.0	3.2	3.4	50	60	70	520.0	522.5	525.0	IF=350mA
TOP-3GM1	3.1	3.3	3.5	100	120	140	520.0	522.5	525.0	IF=700mA

Features
- High brightness, high performance
- High color purity, good uniformity
- Low light decline, long working life

Application
- Generial Lighting

TOP (1W/3W)

BLUE (InGaN)

Part Number	Forward Voltage (V)			Luminous Flux (Lm)			Dominant wavelength (nm)			Test Condition
	Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
TOP-1BD1	3.0	3.2	3.4	10	17	25	460.0	462.5	465.0	IF=350mA
TOP-3BE3	3.1	3.3	3.5	20	35	50	460.0	462.5	465.0	IF=700mA

Features
- High brightness, high performance
- High color purity, good uniformity
- Low light decline, long working life

Application
- Generial Lighting

TOP (1W/3W)

WHITE (InGaN)

Part Number	Forward Voltage (V)			Luminous Flux (Lm)			Dominant wavelength (nm)			Test Condition
	Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
TOP-1BD1	3.0	3.2	3.4	100	120	140	6000	7000	8000	IF=350mA
TOP-3BE3	3.1	3.3	3.5	200	220	240	6000	7000	8000	IF=700mA

Features
- High brightness, high performance
- High color purity, good uniformity
- Low light decline, long working life

Application
- Generial Lighting

Coming soon

Coming soon

Lamp LED

Color

Code	Emitting Color and Material	Wavelength (nm)
UR	Ultra Red (AlGaInP)	635
UE	Ultra Orange (AlGaInP)	625
SE	Super Orange (AlGaInP)	615
UA	Ultra Amber (AlGaInP)	605
UY	Ultra Yellow (AlGaInP) (AlGaInP)	588
UG	Ultra Yellow Green (AlGaInP)	572
PG	Pure Green (INGaN)	515-530
BG	Bluish Green (INGaN)	500-515
UB	Blue (INGaN)	470
PR	Pure Red (GaAsP/GaP)	700(λp)
SR	Super Red (GaAsP/GaP)	640
HE	High Orange (GaAsP/GaP)	620
HA	High Amber (GaAsP/GaP)	610
HY	High Yellow (GaAsP/GaP)	585
HG	High Yellow Green (GaAsP/GaP)	565
SG	Super Yellow Green	560
HR	High Red	660

Code	Emitting Color	Chromaticity Coordinate
BW	Bluel White	X=0.26 Y=0.27
CW	Cool White	X=0.28 Y=0.29
SW	Snow White	X=0.30 Y=0.31
YW	Yellow White	X=0.33 Y=0.34
WW	Warm White	X=0.43 Y= 0.41
PP	Purple Pink	Special Color
PK	Reddish Pink	Special Color
VB	Pueplish Blue	Special Color
GB	Greenish Blue	Special Color
UV	Violet	Special Color

Dual Color

Code	Emitting Colors
RB	Red and Blue
RG	Red and Green
RY	Red and Yellow
OB	Orange and Blue
OG	Orange and Green
YG	Yellow and Green
YB	Yellow and Blue
GR	Green and Red
GO	Green and Orange
GY	Green and Yellow
BR	Blue and Red
BG	Blue and Green
BY	Blue and Yellow
3A	Red Green Blue
3B	Red green Yellow

Drive Mode

Code	Meaning	Sign
A	Common Anode	NP/NP
C	Common Cathode	PN/PN
R	Reverse Electrode	PN/NP

Meaning

Code	Meaning	Notes
None	Normal cutting and package	
A	Stopper type	
B	Normal tape package	Tape to bottom of component is 9mm
C	Normal tape package	Tape to bottom of component is 11mm
D	Normal tape package	Tape to bottom of component is 12mm
E	Normal tape package	Tape to bottom of component is 13mm
F	Normal tape package	Tape to bottom of component is 15mm
G	Right-angle bend tape	Tape to bottom of component is 11mm, Bend to bottom of component is 4mm
H	Right-angle bend tape	Tape to bottom of component is 11mm, Bend to bottom of component is 5mm
K	Right-angle bend tape	Tape to bottom of component is 10mm, Bend to bottom of component is 5mm
L	Append negative Zener	applies to P/N LED chips
M	Append bidirectional Zener	applies to P/N or N/P LED chips
N	Append positive Zener	applies to N/P LED chips
P	Append LED Blinking IC	Blinking frequency: 2.4Hz,1/4 duty cycle
Q	Append LED Blinking IC	Blinking frequency: 1.2Hz,1/4 duty cycle
R	Tape & Reel	

Example

T O L – 5 0 2 S W V C – A

Lens Dimension

Code	Dimension
1	Ø 1.8 mm
2	Ø 2.0 mm
3	Ø 3.0 mm
4	Ø 4.0 mm
5	Ø 5.0 mm
8	Ø 8.0 mm
A	Ø 10 mm
B	2 x 3 mm
C	2 x 5 mm
D	2.1 x 6.9 mm

View Angle

Code	View Angle
1	8-12
2	12-25
3	25-35
4	35-45
5	45-55
6	55-65
7	65-75
9	75-85
A	85-95
B	95-105
C	105-115
D	115-145
E	145-180
T	110/45(Oval)
R	50/30(Oval)

Lens Color

Code	Lens Color
C	Water Clear
W	White Diffused
T	Color Transparent
D	Color Diffused
B	Dark Blue
K	Special Package Color

Lens Shape

Code	Lens Shape
0	Round convex Indicator with Flange
1	Round convex Indicator without Flange
2	Round flat Indicator with Flange
3	Round flat Indicator without Flange
4	Round concave Indicator with Flange
5	Round concave Indicator without Flange
6	Horizontal oval Indicator
7	Vertical oval Indicator
8	Square Indicator with Flange
9	Square Indicator without Flange
A	Triangle Indicator
B	Straw Hat

Typical Luminous Intensity

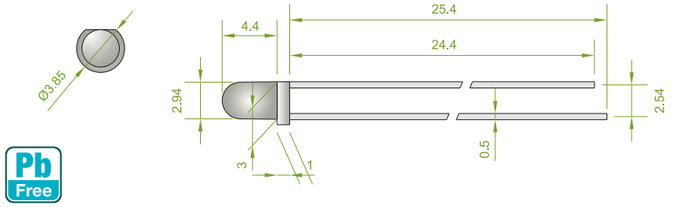
Code	Typical Luminous Intensity (mcd)
1	1-7
2	7-15
3	15-25
4	25-35
5	35-50
6	50-65
7	65-85
8	85-110
9	110-145
A	145-190
B	190-250
C	250-330
D	330-430
E	430-560
F	560-730
G	730-950
H	950-1250
J	1250-1625
K	1625-2100
L	2100-2750
M	2750-3575
N	3575-4650
O	4650-6050
P	6050-7900
Q	7900-10300
R	10300-13400
S	13400-17500
T	17500-22800
U	22800-30000
V	30000-39000

Lamp Type

Code	Lamp Type
L	LED Lamp
LF	Flash LED Lamp
IR	IR LED Lamp
PS	Photo Diode

TOL-30

3mm Round



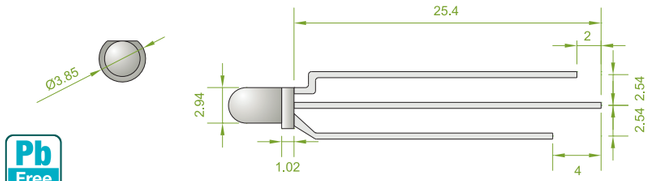
RoHS Pb Free

Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

Application
- Residential lighting
- Automotive interior lighting
- Highway variable message signs
- Medical and industrial equipments
- Consumer and gaming application
- Home appliance lighting application

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-302URKC	AlGaInP	Red	630	Water Clear	1.9	2.5	1000	2000	18
TOL-302UEKC	AlGaInP	Orange	623		2.0	2.4	1000	2000	
TOL-302UANC	AlGaInP	Amber	605		2.0	2.4	2000	4000	
TOL-302UYMC	AlGaInP	Yellow	590		2.0	2.3	1500	3000	
TOL-302UGFC	AlGaInP	Green	573		2.0	2.4	350	700	

3mm Round (Bi-Color)



RoHS Pb Free

Luminous Intensity measurement allowance is ±15%.
Wave length: the typical accuracy is ±2nm.

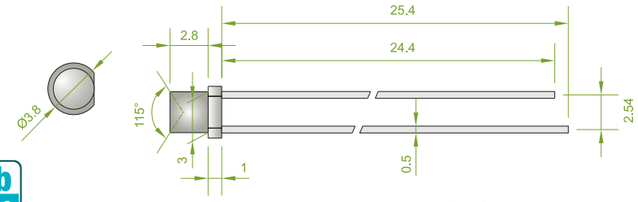
Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

Application
- Residential lighting
- Automotive interior lighting
- Highway variable message signs
- Medical and industrial equipments
- Consumer and gaming application
- Home appliance lighting application

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-307RGAW	GaAlAs/GaAs	Red	640	White Diffused	1.8	2.1	34	80	60
	GaP/GaP	Green	570		2.2	2.5	15	30	
TOL-307RBAW	AlGaAs/AlGaAs	Red	643		2.0	2.4	65	130	60
	InGaN/GaN	Blue	470		3.0	4.0	110	220	70

TOL-34

3mm Concave



RoHS Pb Free

Luminous Intensity measurement allowance is ±15%.
Wave length: the typical accuracy is ±2nm.

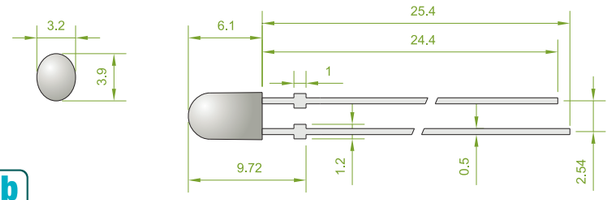
Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

Application
- Home appliance & lighting
- Automotive interior lighting
- Highway variable message signs
- Medical and industrial equipments
- Consumer and gaming electronics application

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-34BSR4C	AlGaInP	Red	640	Water Clear	2.0	2.5	17	35	105
TOL-34EUYBC	AlGaInP	Yellow	590		2.0	2.5	125	250	150
TOL-34BUY4C	AlGaInP	Green	574		2.0	2.4	20	40	105

TOL-36

3mm Oval



RoHS Pb Free

Luminous Intensity measurement allowance is ±15%.
Wave length: the typical accuracy is ±2nm.

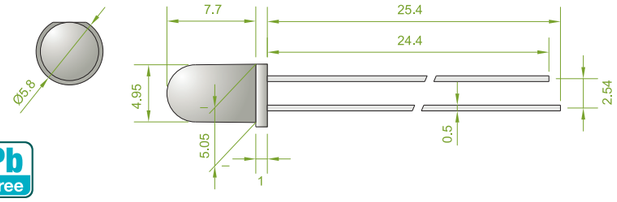
Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

Application
- Full colour video screens
- Variable Messasge Sign (VMS)
- Outdoor advertising boards

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-36TUEHK-A	AlGaInP	Orange	625	Color Diffused	1.6	2.8	600	1200	110/50
TOL-36TUYJK-A	AlGaInP	Yellow	592		1.6	2.8	750	1500	
TOL-36TUGJK-A	InGaN	Pure Green	525		3.2	3.8	1000	1500	
TOL-36TUBEK-A	InGaN	Blue	470		3.2	3.8	225	450	

TOL-50

5mm Round



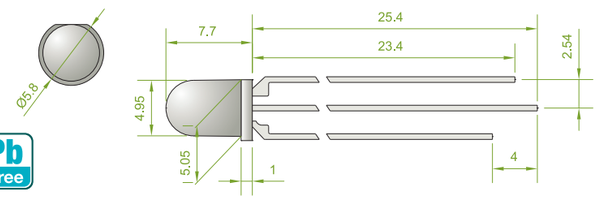
RoHS Pb Free

Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

Application
- Traffic signs
- Residential lighting
- Automotive interior lighting
- Highway variable message signs
- Medical and industrial equipments
- Consumer and gaming application
- Home appliance lighting application

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-502URNC	AlGaInP	Red	630	Water Clear	2.0	2.4	2000	4000	15
TOL-502UELC	AlGaInP	Orange	623		2.0	2.5	1200	2400	
TOL-502UANC	AlGaInP	Amber	605		2.0	2.5	2400	4500	
TOL-502UYMC	AlGaInP	Yellow	590		2.0	2.3	1550	3000	
TOL-502UGJC	AlGaInP	Green	572		2.0	2.5	750	1500	
TOL-503UREC	AlGaInP	Red	630		2.0	2.4	250	500	30
TOL-503UEHC	AlGaInP	Orange	623		2.0	2.4	600	1200	
TOL-503UAHC	AlGaInP	Amber	605		2.0	2.5	500	1000	
TOL-503UYFC	AlGaInP	Yellow	590		2.0	2.5	360	730	
TOL-504URBC	AlGaInP	Red	630		2.0	2.4	100	200	40
TOL-504UYHC	AlGaInP	Yellow	590		2.0	2.4	500	1000	45

5mm Round (Bi-Color)



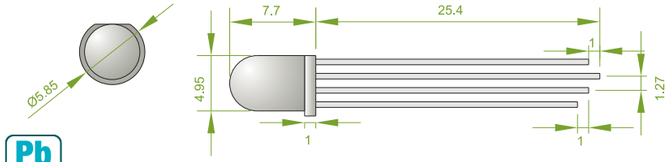
RoHS Pb Free

Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

Application
- Traffic Signs
- Home appliance & lighting
- Automotive interior lighting
- Highway Variable Message Signs
- Medical and Industrial Equipments
- Consumer and Gaming Electronics application

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-506RGAW	GaAlAs/GaAs	Red	640	White Diffused	1.8	2.1	42	95	60
	GaP/GaP	Green	570		2.2	2.5	14	30	
TOL-509RBAW	AlGaAs/AlGaAs	Red	643		2.0	2.4	60	120	70
	InGaN/GaN	Blue	470		3.0	4.0	100	200	80

5mm Round (RGB)



RoHS Pb Free

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Wave length: the typical accuracy is ±2nm.

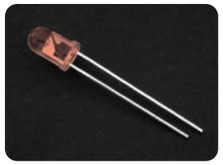
Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

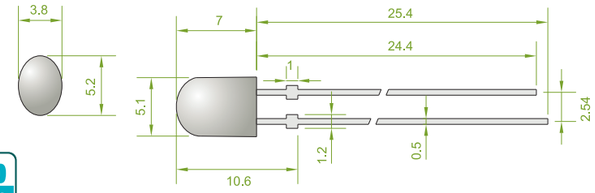
Application
- Traffic signs
- Residential lighting
- Automotive interior lighting
- Highway variable message signs
- Medical and industrial equipments
- Consumer and gaming application
- Home appliance lighting application

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-5043ACC	AlGaInP	Orange	624	Water Clear	2.0	2.6	200	400	50
	InGaN	Green	525		3.5	4.0	1250	2500	35
	InGaN	Blue	465		3.5	4.0	500	1000	40
TOL-5043AAC	GaAlAs/GaAs	Red	640		2.0	2.4	103	190	55
	InGaN/Sic	Green	525		3.5	4.0	980	2000	35
	InGaN/GaN	Blue	470		3.5	4.0	390	750	35

TOL-56

5mm Oval





RoHS Pb Free

Luminous Intensity measurement allowance is ±15%.
Wave length: the typical accuracy is ±2nm.

Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

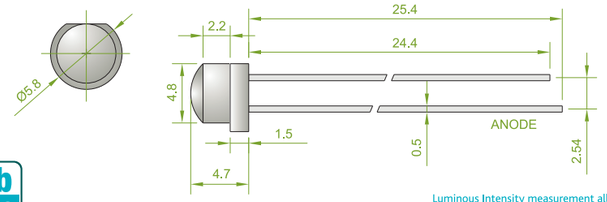
Application
- Full colour video screens
- Variable Messasge Sign (VMS)
- Outdoor advertising boards

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-56TUEJK-A	AlGaInP	Orange	625	Color Diffused	1.6	2.8	700	1400	110/45
TOL-56TUYHK-A	InAlGaP	Yellow	592		1.9	2.5	600	1200	
TOL-56TPGKK-A	InGaN	Pure Green	525		2.6	3.8	900	1800	
TOL-56TUBFK-A	InGaN	Blue	470		2.6	3.8	350	700	

LAMP

TOL-4B

4.8mm



Luminous Intensity measurement allowance is ±15%.
Wave length: the typical accuracy is ±2nm.

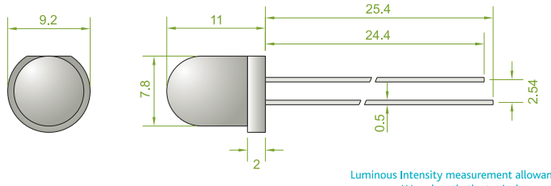
Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

Application
- Full colour video screens
- Variable Messasge Sign (VMS)
- Outdoor advertising boards

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-4BEUE9C	AlGaInP	Orange	623	Water Clear	2.0	2.4	65	130	150
TOL-4BDUYEC	AlGaInP	Yellow	590		2.0	2.3	225	450	120
TOL-4B7UG9C	AlGaInP	Green	572		2.0	2.6	70	140	70
TOL-4B7UBCC	InGaN	Blue	470		3.0	3.5	160	300	70

TOL-80

8.0mm Round



Luminous Intensity measurement allowance is ±15%.
Wave length: the typical accuracy is ±2nm.

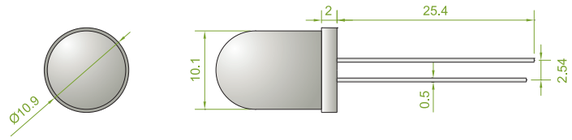
Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

Application
- Indicator light
- Medical and industrial equipments
- Consumer and gaming application

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-803URDC	AlGaInP	Red	640	Water Clear	2.0	2.4	200	400	30
TOL-803HYBC	GaAsP	Yellow	590		2.1	2.4	100	200	
TOL-803HGBC	GaP/GaP	Green	525		2.2	2.5	100	200	
TOL-803UBNC	InGaN	Blue	470		3.5	4.0	1700	3600	

TOL-A03

10mm Round



Luminous Intensity measurement allowance is ±15%.
Wave length: the typical accuracy is ±2nm.

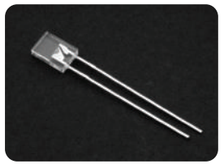
Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

Application
- Indicator light
- Medical and industrial equipments
- Consumer and gaming application

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-A03URHC	AlGaInP	Red	630	Water Clear	2.0	2.5	500	1100	30
TOL-A03UEOC	AlGaInP	Orange	623		2.1	2.6	2500	5000	
TOL-A03UYOC	AlGaInP	Yellow	590		2.2	2.5	2600	5200	
TOL-A03UBNC	InGaN/SiC	Blue	470		3.5	4.0	2000	4000	

TOL-C9

Rectangular (2.05mm x 5.02mm)



Luminous Intensity measurement allowance is ±15%.
Wave length: the typical accuracy is ±2nm.

Features
- Superior brightness
- Low power consumption
- Reliability life test confirmed
- 2kV ESD withstanded

Application
- Automotive interior lighting
- Medical and industrial equipments
- Consumer and gaming application
- Home appliance lighting application

Part Number	Chip			Lens Color	V_F(V) at I_F=20mA		I_V(mcd) at I_F=20mA		Viewing Angle 2θ½
	Material	Emitted Color	λ_d(nm)		TYP.	MAX.	MIN.	TYP.	
TOL-C9EUR3K	AlGaInP	Red	640	Water Clear	2.0	2.5	7.0	15	170
TOL-C9EUE3K	AlGaInP	Orange	623		2.0	2.4	10	20	165
TOL-C9EHY1D	GaAsP/GaP	Yellow	590		2.0	2.4	3.0	6.0	160
TOL-C9BHG2K	GaP/GaP	Green	568		2.2	2.5	5.0	10	100

LAMP

coming soon